ASSIGNMENT-13 : CLOUD COMPUTING BASIC

GUIDED BY: VIPUL MISTRI

Q.1 Create a comparison table listing at least 3 differences each between IaaS, PaaS, and SaaS, and give one real-world app example for each type (e.g., Spotify, Flipkart, etc.).

ANS : Here is a comparison table outlining the key differences between Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS):

Feature / Aspect	**IaaS (Infrastructure as a Service)**	**PaaS (Platform as a Service)**	**SaaS (Software as a Service)**
Control & Management	You manage the OS, runtime, middleware, data, and applications; vendor manages virtualization, storage, network, and servers.	You manage the application code and data; vendor manages the OS, runtime, middleware, and infrastructure.	Vendor manages everything (application, runtime, OS, servers, storage, network, and security).
Primary Target Audience	Network architects, systems administrators, and DevOps engineers needing granular infrastructure control.	Developers and software engineers focusing on building and deploying applications without server management.	End users, consumers, and business employees accessing ready-to-use software applications.
Flexibility vs. Convenience	Maximum flexibility and customization, but requires high setup time and ongoing system administration.	Balanced flexibility; streamlined application development and deployment with	Maximum convenience and accessibility; zero maintenance or setup required, but limited customization.

ASSIGNMENT-13 : CLOUD COMPUTING BASIC

GUIDED BY: VIPUL MISTRI

Feature / Aspect	**IaaS (Infrastructure as a Service)**	**PaaS (Platform as a Service)**	**SaaS (Software as a Service)**
		zero OS overhead.	
Real-World App Example	**Amazon EC2 (AWS) / Google Compute Engine**	**AWS Elastic Beanstalk / Heroku**	**Spotify / Flipkart**

Q.2 Pick any two apps you use daily (such as Zomato and Instagram) and explain in 2-3 lines how they might use cloud computing in their backend operations.

ANS : Here is how two popular everyday apps leverage cloud computing in their backend operations:

1. **Zomato:** Zomato relies heavily on cloud computing (such as AWS infrastructure) to dynamically scale its backend servers during peak meal-ordering hours, handle millions of real-time GPS delivery tracking requests, and process distributed database transactions connecting restaurants, delivery partners, and customers instantly.

2. **Instagram:** Instagram uses massive cloud data centers and content delivery networks (CDNs) to store, transcode, and deliver high-resolution photos and videos globally, while running cloud-based machine learning algorithms to personalize feeds, recommend Reels, and moderate content in real time.

ASSIGNMENT-13 : CLOUD COMPUTING BASIC

GUIDED BY: VIPUL MISTRI

Q.3 Sign up for an AWS Free Tier account (use your own email/phone), log in to the AWS Management Console, and take a screenshot of your dashboard showing your account name.

ANS :

How to Sign Up for an AWS Free Tier Account

1. **Navigate to the AWS Homepage:**
    - Go to the official AWS website: aws.amazon.com
2. **Start Registration:**
    - Click the **Create an AWS Account** (or **Sign Up**) button in the top right corner.
3. **Enter Account Details:**
    - Provide your email address, choose a secure password, and enter your AWS account name.
4. **Contact & Billing Information:**
    - Enter your contact information and a valid credit/debit card for identity verification (AWS requires this to prevent fraud, but you will not be charged for services covered under the Free Tier limits).
5. **Verify Your Identity:**
    - Complete the phone verification step (via SMS or voice call) by entering the PIN code displayed on your screen.
6. **Select a Support Plan:**
    - Choose the **Basic Support (Free)** plan to complete registration.

Finding Your Account Name on the AWS Management Console Dashboard

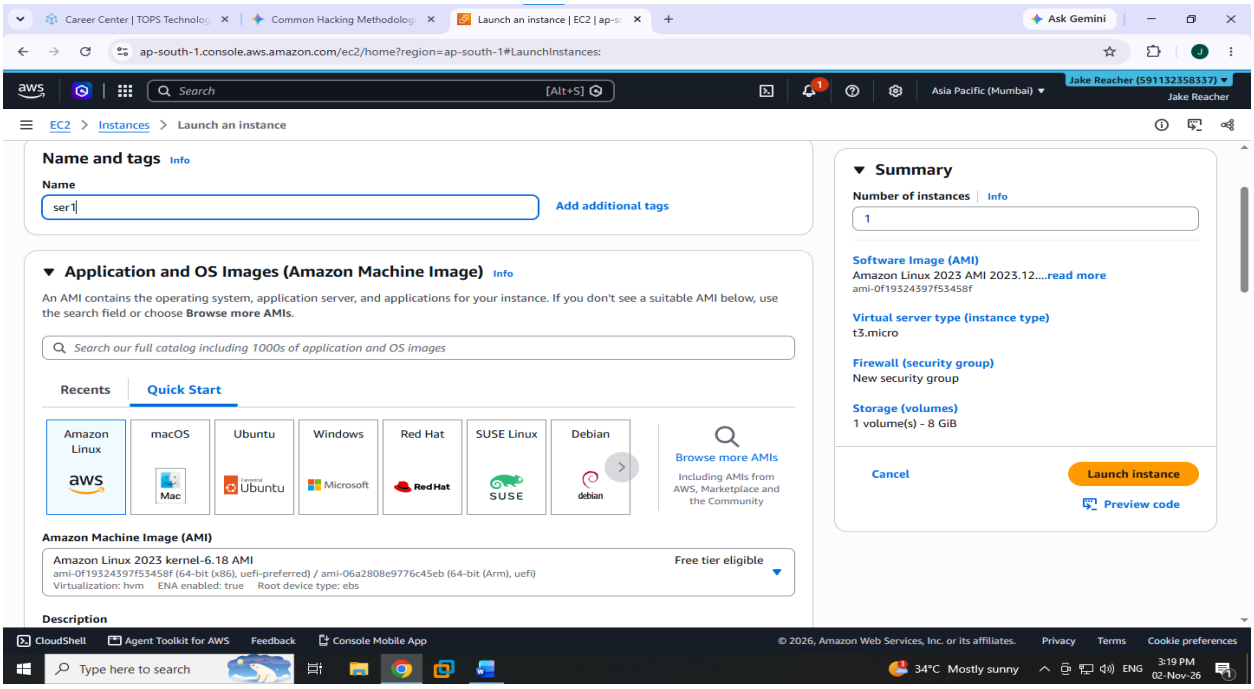
1. Once logged in, you will land on the **AWS Management Console** home dashboard.
2. Look at the **top-right corner** of the navigation bar.
3. Your **Account Name** (or the alias you assigned during setup) is displayed directly next to the region selector and notification bell icon (often formatted with your account ID number in parentheses.

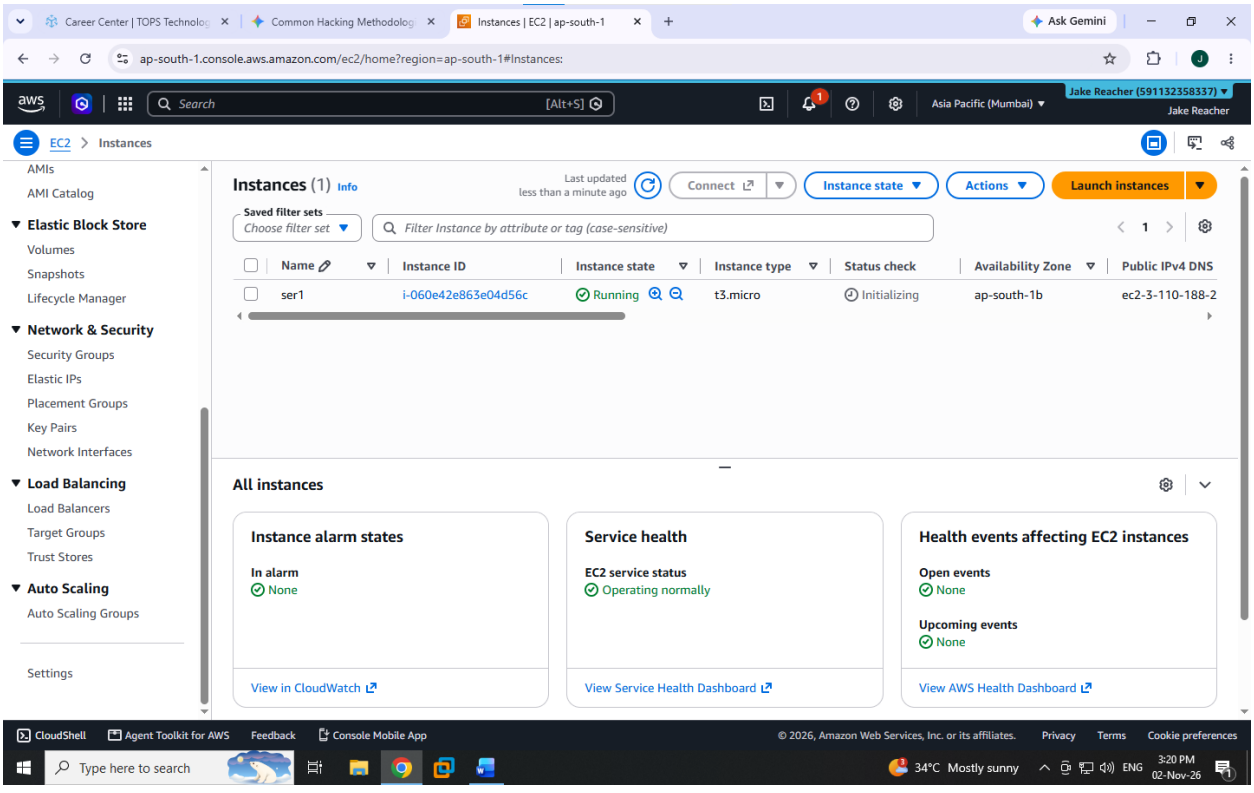
ASSIGNMENT-13 : CLOUD COMPUTING BASIC

GUIDED BY: VIPUL MISTRI

Q.4Launch a basic EC2 instance on AWS using the Free Tier (choose t2.micro, Amazon Linux 2), and once it is running, take a screenshot of the instance details page showing its status.

ANS :





ASSIGNMENT-13 : CLOUD COMPUTING BASIC

GUIDED BY: VIPUL MISTRI

Q.5 After launching your EC2 instance, stop it from the AWS console and note the changes in its status and billing estimate. Hint: Check the 'Instance State' and 'Estimated charges' sections in the AWS console.

ANS :

